

Carlos José Requena Jiménez

Solutions Architect | Enterprise Architecture & Cloud Strategy

Palma, Spain • +34 669 258 306 • cjrequena@outlook.es

linkedin.com/in/cjrequena • github.com/cjrequena • cjrequena.com • medium.com/@cjrequena • twitter.com/cjrequena

Solutions Architect with 15+ years in enterprise environments, working across the full architectural stack — from event-driven distributed systems and multi-tenant platform design to AWS environments defined entirely as code across a multi-account organization. Equally comfortable setting the standard and writing the reference implementation: API contracts, messaging semantics, deployment topology, identity and authorization models, and the CI/CD pipelines that carry them to production.

Core stack — Java 21 / Spring Boot 3.5 • AWS (ECS Fargate, RDS PostgreSQL, MSK, Cognito) • Terraform • Apache Kafka • PostgreSQL • TypeScript / Next.js • GitLab CI • Event Sourcing / CQRS • Multi-Tenant SaaS

Professional Experience

TUI

Solutions Architect • Jan, 2017 — Present

- Defined architecture strategy for the global sourcing and integration platforms behind multi-regional travel operations, sizing for high-demand seasonal workloads and setting the scalability and resilience targets delivery teams build against — 99.99% uptime and a 40% improvement in peak-load scalability across business-critical services
- Led the decomposition of a monolithic booking platform into a set of independently deployable, reactive Java/Spring Boot services with event-driven integration — service boundaries, ownership model, deployment topology, and automated release paths — cutting deployment time by 60% and improving delivery throughput, while consolidating over a dozen regional brand storefronts onto a single whitelabel frontend
- Established asynchronous messaging standards on Kafka and SNS/SQS — event contracts, consumer isolation, failure handling, and replay — removing synchronous coupling between domains and containing the blast radius of downstream outages; applied this to federate real-time availability search across multiple external and internal suppliers, streaming results to clients as they arrive rather than blocking on the slowest supplier
- Introduced CQRS as the persistence model for the ancillary booking-lifecycle domain — an independently scaled command service handling writes and a query service serving reads from a document store, with domain events typed and versioned through a dedicated schema registry and AsyncAPI contracts — letting read and write paths scale independently while keeping downstream consumers aligned on event shape across versions
- Owned architecture governance adopted across multiple global teams: design reviews, security and development standards, reusable integration patterns, and versioned REST contracts with backward-compatibility rules that let consuming teams evolve independently of provider release cycles — with the application landscape and its dependencies tracked in LeanIX
- Defined the API gateway strategy for externally consumed services on Apigee — OAuth2 token verification and CORS policy standardized across the proxy layer so internal services never expose public endpoints directly — and led the phased migration of the proxy estate from Apigee Edge to Apigee X across test, pre-prod, and production for consumers spanning partner integrations and the mobile app team
- Influenced technical direction across Product, Engineering, Security, and DevOps, translating enterprise strategy into concrete platform decisions and making their operational and security implications explicit to non-technical stakeholders
- Mentored engineers and architects through design reviews, pairing on architectural decisions, and written guidance

Environment: Java 17/21 • Spring Boot 3.x (WebFlux) • AWS ECS Fargate • RDS PostgreSQL • DocumentDB • MSK / Apache Kafka • SNS/SQS • Cognito • Apigee (Edge → X migration) • Terraform • GitLab CI • REST APIs • LeanIX

Hotelbeds

Application Architect • Sep, 2015 — Jan, 2017

- Designed and scaled backend services processing millions of daily transactions across high-availability travel distribution platforms — sized for peak distribution load, disaster recovery, and data-integrity guarantees
- Led modernization of legacy systems into modular microservices — service decomposition, data ownership, and integration contracts — across the core travel distribution platform

- Improved platform performance by up to 30% by targeting structural bottlenecks — call topology, data access paths, and caching strategy — rather than isolated code tuning

Environment: Java • Spring • REST APIs • High Availability

Selected Work

Jaspe

Architect & Primary Developer • Personal platform project

Multi-tenant SaaS booking and business-management platform for service businesses — end-to-end reference implementation: seven-service backend, three Next.js frontends, and AWS infrastructure as Terraform.

- Architected a seven-service backend across three deliberate styles chosen per bounded context — Hexagonal (business, customer), CQRS + Event Sourcing (booking command/query handlers, backed by a custom event-store library published as a versioned artifact to a private Maven registry), and Layered (auth, payment, notification) — integrated via synchronous REST and CloudEvents-typed Kafka events with per-consumer DLQs and replay
- Enforced correctness in PostgreSQL rather than application code: GiST exclusion constraints over `daterange` / `tstzrange` (with `btree_gist` for UUID/enum equality) make overlapping location hours, staff schedules, and asset bookings unrepresentable, alongside Row-Level Security tenant isolation rolled out across every service and pinned with per-service regression tests
- Owned the identity and authorization boundary end-to-end: Cognito User Pool with three Lambda triggers (post-confirmation provisioning, pre-token-generation claim enrichment, pre-signup federation), a Cognito JWT authorizer at the API Gateway edge, and claim-based method authorization inside each service
- Defined the delivery topology as Terraform (API Gateway → VPC Link → internal ALB → ECS Fargate, multi-account, eu-west-1) shipped through GitLab CI parent/child pipelines gated on per-service change detection, and delivered three Next.js frontends (business, platform-admin, patient-facing) sharing one backend through a Stripe Connect payment integration
- Directed the majority of implementation through Claude Code and Kiro as AI coding agents rather than hand-writing it — authoring the machine-readable architecture contracts (workflow, model-routing, naming/wire-format, git conventions) that let agents work correctly without step-by-step supervision, plus Kiro hooks that auto-detect architecture-documentation drift on every commit — with over half of all commits across the platform AI-co-authored and every agent-produced change reviewed against the reference architecture before merge

Environment: Java 21 • Spring Boot 3.5 • PostgreSQL (RLS, GiST) • Apache Kafka (CloudEvents) • AWS (ECS Fargate, API Gateway, Cognito, Lambda) • Terraform • Next.js / TypeScript • Stripe Connect • GitLab CI • Hexagonal • CQRS / Event Sourcing • Multi-Tenant SaaS • Claude Code / Kiro (AI-Agent-Directed Engineering)

Technical Expertise

Architecture & Design

Microservices • Hexagonal Architecture (Ports & Adapters) • CQRS • Event Sourcing • Domain-Driven Design • Event-Driven Architecture • API-First & Versioned REST Design • Multi-Tenancy & Tenant Isolation • Saga / Compensating Transactions • Idempotency & Optimistic Concurrency • High Availability, Resilience & Graceful Degradation

AWS, Infrastructure & Delivery

ECS Fargate • API Gateway • Lambda • RDS PostgreSQL & RDS Proxy • ElastiCache Redis • MSK / Managed Kafka • Cognito (OIDC/JWT, custom claims, M2M client credentials) • S3 • Secrets Manager • CloudWatch • VPC & Network Segmentation • ALB & Auto Scaling • Organizations, SCPs & IAM Identity Center • Multi-Account Landing Zones • Terraform (reusable modules, thin per-environment root stacks, remote-state composition) • Docker • Ansible • GitLab CI (parent/child pipelines, path-based gating, DAG orchestration) • OIDC Keyless Cloud Authentication • Promotion Gates & Trunk-Based Development

Engineering, Data & Security

Java 21 • Spring Boot 3.5 (Cloud Stream, Security / OAuth2 Resource Server, Data JPA) • Python • TypeScript • Next.js (App Router, SSR / RSC) • React • OpenAPI • Micrometer & Distributed Tracing • PostgreSQL (Row-Level Security, GiST exclusion constraints, PostGIS, transaction-scoped tenant context) • Event Store Design (append-only log, snapshots, durable subscription offsets) • Apache Kafka • CloudEvents • Dead-Letter Queues & Replay • Flyway • OAuth 2.0 / OIDC • RBAC & Claim-Based Authorization • Least-Privilege IAM • GDPR / EU Regulatory Assessment • JUnit 5 • Mockito • Testcontainers • Property-Based Testing • Playwright • Locust • SonarQube • arc42, C4, Mermaid

AI-Assisted & Agentic Engineering

Claude Code (agent-instruction contracts, custom skills, slash-command scaffolds, sub-agent model routing) • Kiro (spec/hook-driven agents, automated architecture-doc-coherence enforcement) • LLM-Based Pair Programming & Code Review • Prompt/Role Engineering for Domain-Specific AI Agents • Agentic Workflow Governance (plan-then-execute, human-in-the-loop review) • Multi-Repo AI-Directed Delivery at Scale

Earlier Experience

Onwhyon

Application Architect • Nov, 2014 — Jun, 2015

Lead architect for Bankia Indicex, a public website-maturity scoring platform: crawler-based extraction feeding parallel enrichment against ~10 third-party APIs (Java, Spring 4, AngularJS, MySQL).

Trentisa

Software Architect • Apr, 2010 — Nov, 2014

Architecture for enterprise banking and SEPA platforms; high-throughput financial data processing with reconciliation and data-integrity constraints under regulatory requirements.

Selected engagements: SEPA (direct debit, mandates, receipts and transfers), Bank of Santander (bulk processing for mission-critical business operations), LeasePlan (profitability and financial calculation studies).

Virtual Desk

Software Developer • Sep, 2009 — Apr, 2010

Distributed architecture solutions, technology selection, and code reviews (ZKOSS, Spring 2.5, Hibernate/JPA, Oracle 10g).

Everis Spain

Software Developer • Jan, 2009 — Sep, 2009

Java application architecture, SOA, Liferay portlets, and data modelling, migration, and warehousing.

OLTP Voice, Caracas

Software Developer • Jan, 2008 — Jan, 2009

Middleware for Movistar post-paid transaction and account activation systems.

Imolko, Caracas

Software Developer • Jan, 2006 — Jan, 2008

Web and business components for SMS-based platforms (Java, Web Services / SOA, SMPP).

Education

EUDE Business School

MBA (Master in Business Administration) • 2010 — 2011

Business strategy, finance, general management, project management, and operations management.

UNEXPO — Universidad Nacional Experimental Politécnica Antonio José de Sucre

Electronic Engineer • Mar, 2005 — Mar, 2009

Telecommunications, digital electronics, computing, industrial controls, and signal processing.

Certifications & Training

CAPSIDE — Advanced Architecting on AWS • 2018 — (48 hours)

Advanced Architecting on AWS

European Scrum — Expert Scrum Master Certified • 2015 — (16 hours)

European Scrum

IBM Training — IBM WebsPhere Lombardi BPM Process Analysis Methods II IBM Training. • 2010 — (48 hours)

Sun Microsystems — Java (TM) Programing Language sl275. • 2006 — (96 hours)

Nationals Instruments — Developing Applications with LabView. • 2006 — (48 hours)

UNEXPO — Advanced Programming Pic's microcontroller 16F877A. • 2004 — (48 hours)

Languages

- Spanish • Native
- English • Full Professional Proficiency